

Assignment 8: Damascene Process

This report explores the conventional metallization process and its limitations, and how the damascene process addresses these challenges.

Question 1: Conventional Metallization Process

This analysis will explore the limitations of conventional copper metallization in integrated circuits, focusing on patterning difficulties, junction spiking, and poor oxidation resistance. We'll identify a specific challenge, examine copper diffusion profiles, and analyse how diffusion temperature affects these profiles. This investigation aims to shed light on the issues with conventional metallization and pave the way for exploring potential solutions.

The conventional metallization process involves depositing a thin film of metal directly onto the surface of a substrate. This process is simple and straightforward, but it has several limitations that make it unsuitable for use in modern ICs.

The default process followed for metallization is shown by figure below,

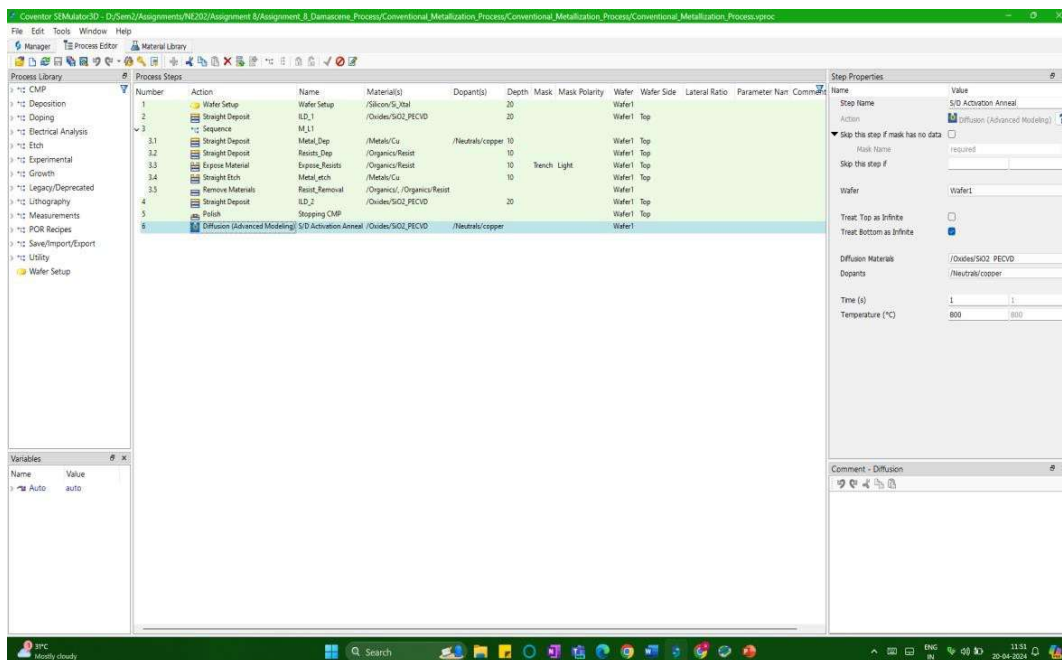


Figure 1 : Process steps for metallization

Results of simulated process flow is as follows,

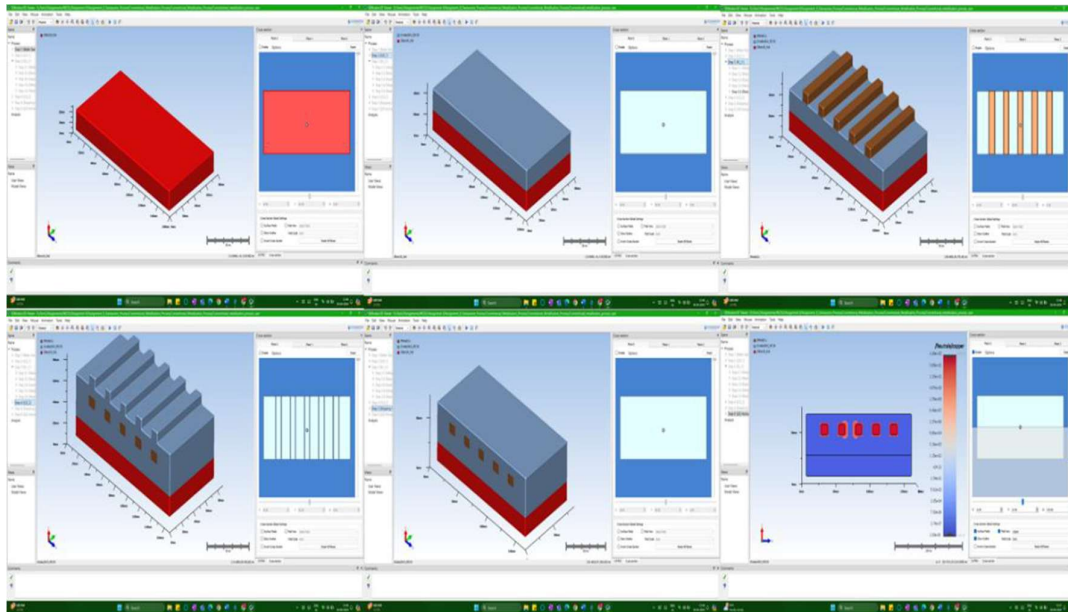


Figure 2 : Simulated results for metallization

Question1(a)

After annealing at 800°C for 1s, Cu has diffused into SiO₂ and has penetrated the junction (Figure 3) so problem we are dealing with is **Junction spiking/Cu poisoning**.

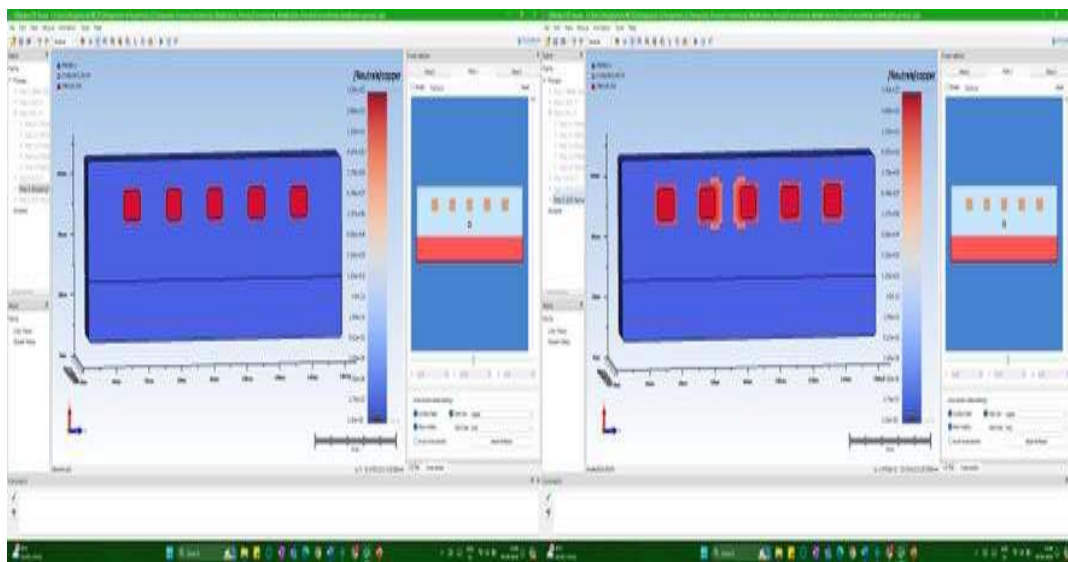


Figure 3 : Cu poisoning after annealing

Question1(b)

After annealing Cu with concentration of $1e15 \text{ cm}^{-3}$ and diffusivity of $3.51214e-15 \text{ cm}^2/\text{sec}$ in SiO_2 should diffuse across the junction, this can be seen in Cu diffusion profile before and after annealing is shown in figure below.

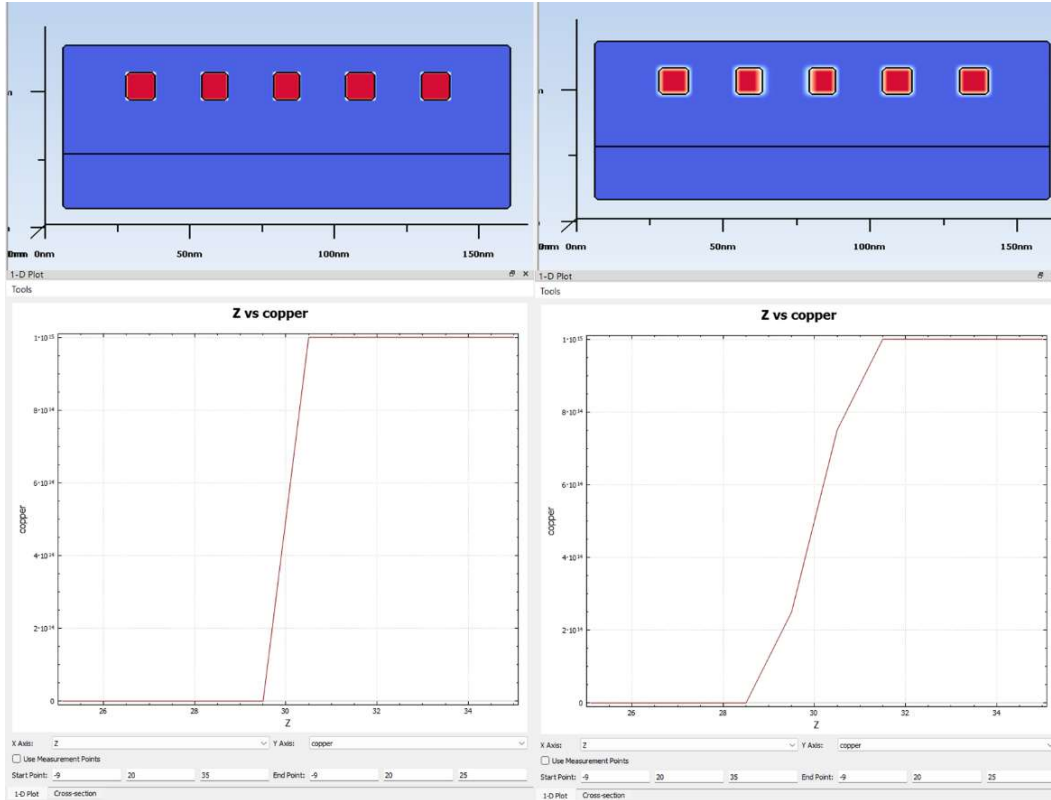


Figure 4 : Cu Profile before and after annealing

We can see the slope at Cu- SiO_2 junction is changed which tells us that there have been Cu poisoning.

Question1(c)

It is observed that as temperature is increased for annealing, diffusivity of Cu in SiO₂ increases (Table 1) and could short the junction.

Table 1 : Diffusivity of Cu

Sl.No.	Temperature (°C)	Diffusivity in SiO ₂ (cm ² /sec)
1.	550	9.49289e-18
2.	750	1.3563e-15
3.	1050	1.39046e-13

The results simulated device shows increase in Cu poisoning with respect to anneal temperature. This is due to exponential dependence of diffusion coefficient on temperature (given by Arrhenius relation).

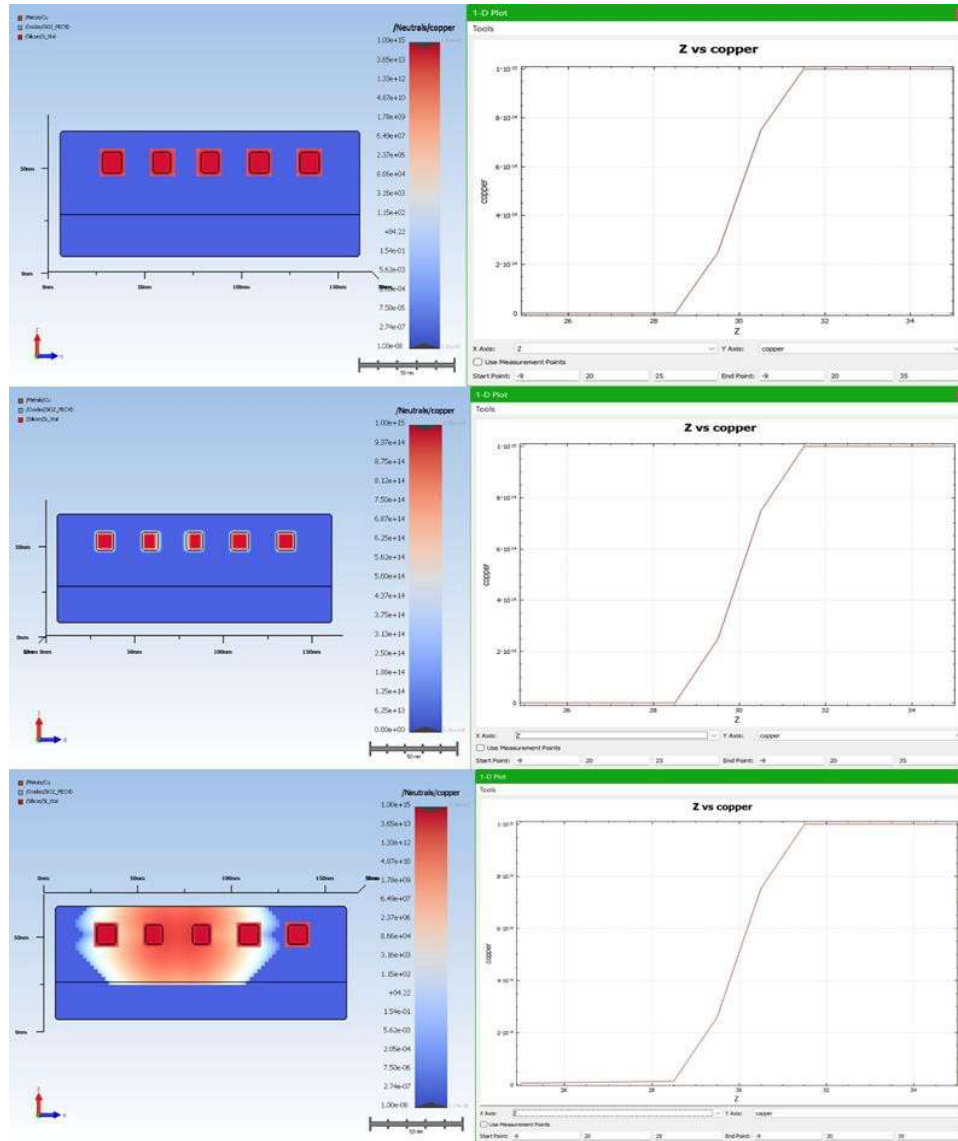


Figure 5 : Cu Profile at 550°C, 750°C and 1050°C

Question 2: Damascene process

The damascene process is used to make metal connect for devices. The process starts with covering the device Inter metal dielectric layers and then etching tiny vias through the insulation layers. after that a protective barrier of TaN and copper seed layer are added, then trench is filled with copper using electroplating. Finally, polishing removes excess copper.

The process flow and simulated results are shown in figure below,

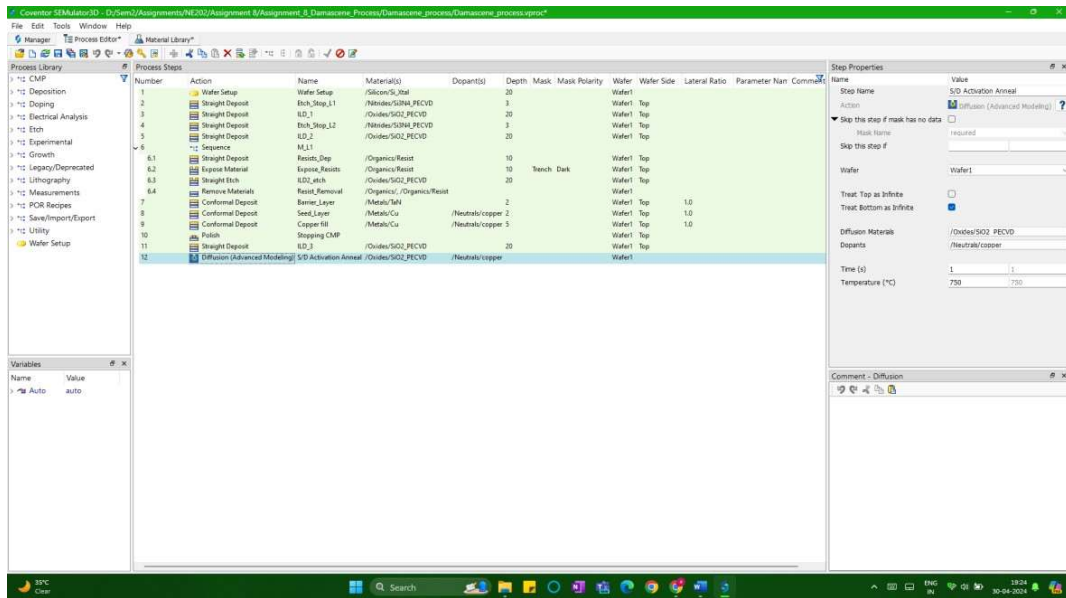


Figure 6 : Process flow for damascene process

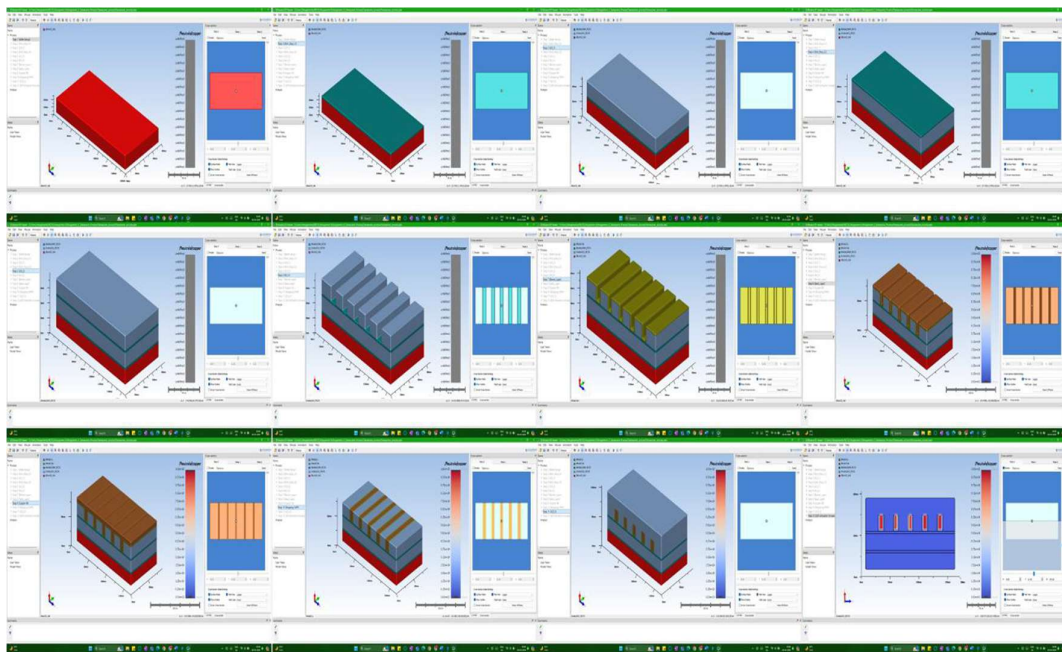


Figure 7 : Simulated results for damascene process

Question2(a)

Changes observed in the process flow of damascene process as compared to conventional method are -

- **Etch selectivity:** Traditional methods etch the metal layer to define the pattern and subsequently fill the remaining areas with ILD. In damascene process selectively removes the ILD, allowing for the deposition of conductive materials like copper, which offer superior conductivity compared to commonly used aluminium in conventional methods.
- **Patterning technique:** Unlike conventional methods damascene process does not rely on wet etching to remove excess metal, it utilizes CMP, a more controlled and gentler polishing technique, for planarization.
- **Multi-layer capability:** Traditional approaches require separate via formation and metal trench filling steps. The single damascene process offers the advantage of creating both vias and trenches simultaneously within the ILD, followed by a single filling step.

Question2(b)

For proper observation anneal time was increased from 1seconds to 10 seconds. For diffusivity calculation of Si_3N_4 , TiN and W parameters were added in Cu material property (Figure 8).

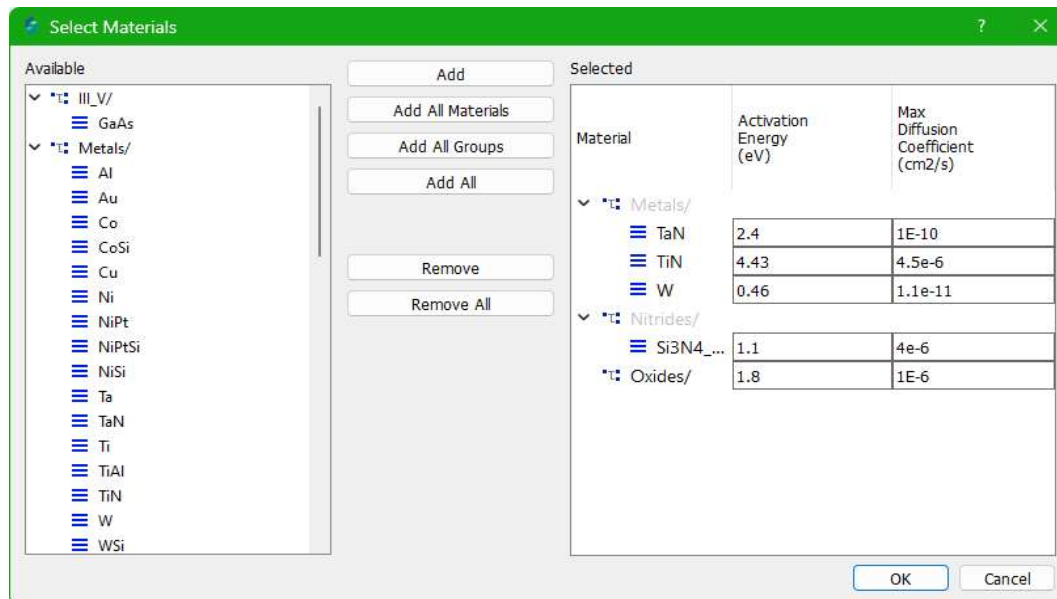


Figure 8 : Diffusion Parameters

Model was run for three temperatures with different diffusion barrier layer and calculated diffusivity is given by Table 2. For default diffusion barrier (TaN) results are shown in Figure 9.

Table 2 : Diffusivity of copper

Sl.No.	Temperature (°C)	TaN (cm ² /sec)	Si ₃ N ₄ (cm ² /sec)	TiN (cm ² /sec)	W (cm ² /sec)	SiO ₂ (cm ² /sec)
1.	550	2.010e-25	7.345e-13	3.352e-33	1.676e-14	9.4928e-18
2.	750	1.5013e-22	1.523e-11	6.744e-28	5.959e-14	1.3563e-15
3.	1050	7.203e-20	2.5808e-10	3.415e-15	1.945e-13	1.3904e-13

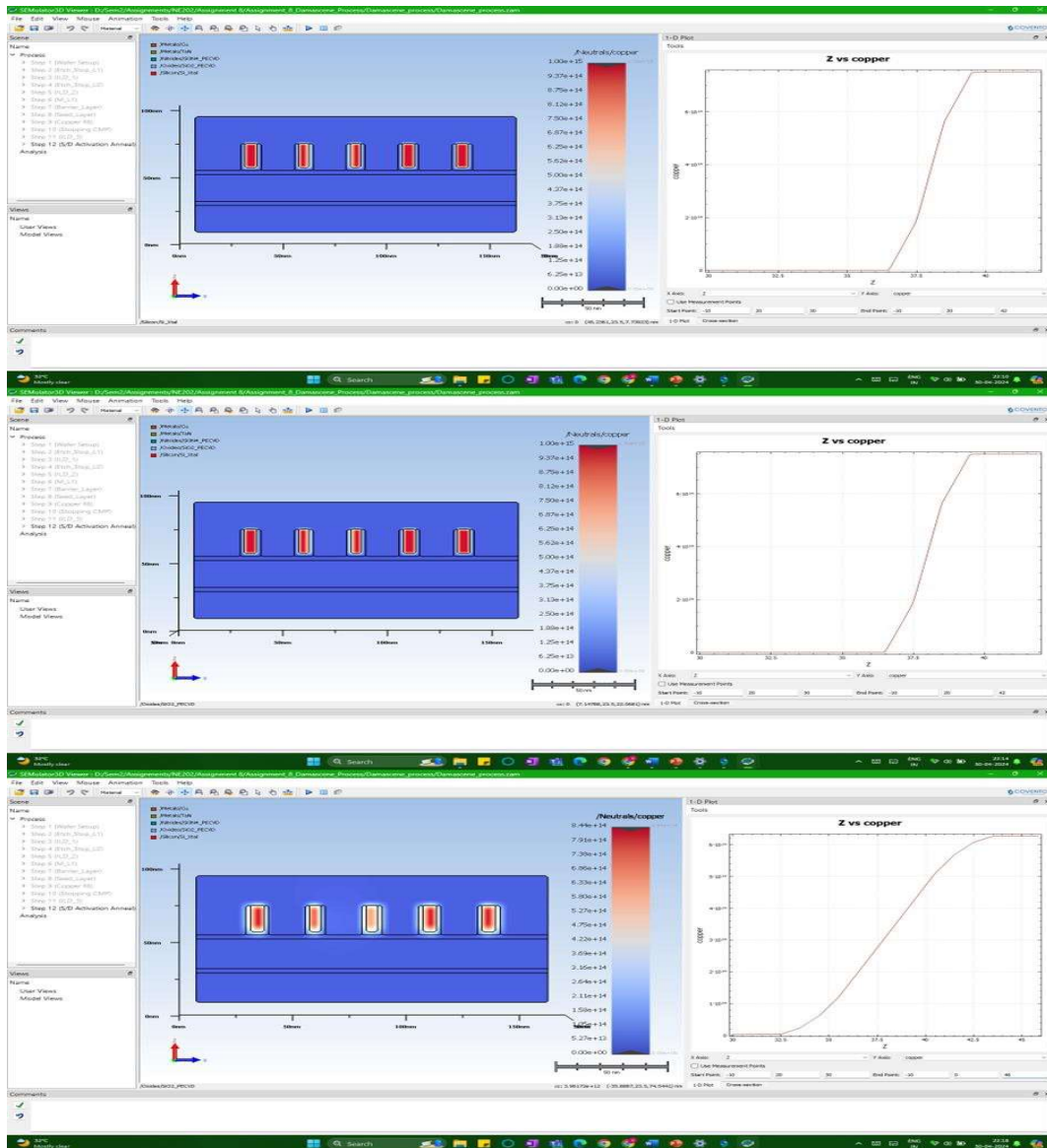


Figure 9 : Cu Profiles for TaN barrier

Simulated results for Si_3N_4 barrier are shown in figure below,

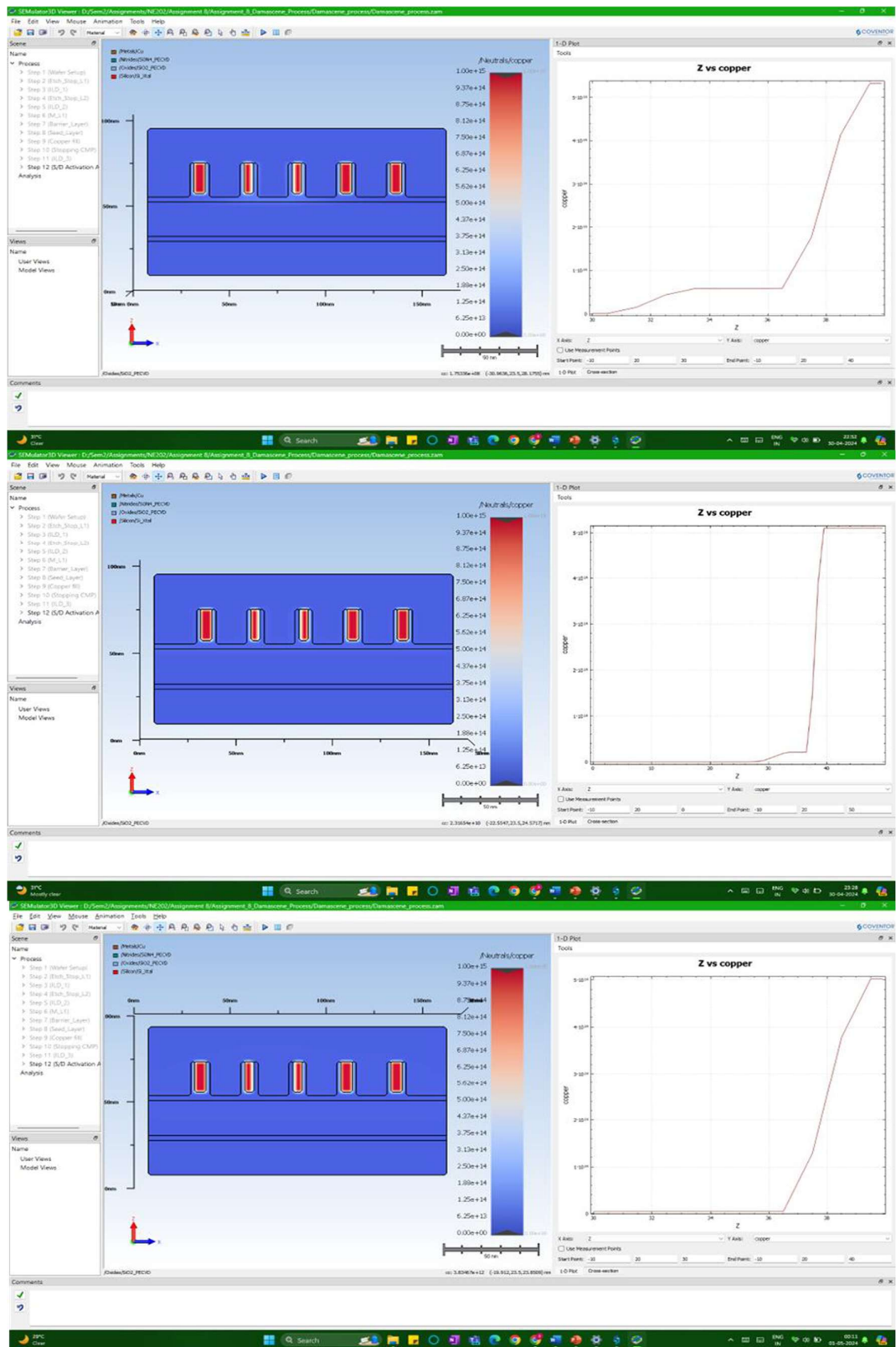


Figure 10 : Cu Profiles for Si_3N_4 barrier

Simulated results for TiN barrier are shown in figure below,

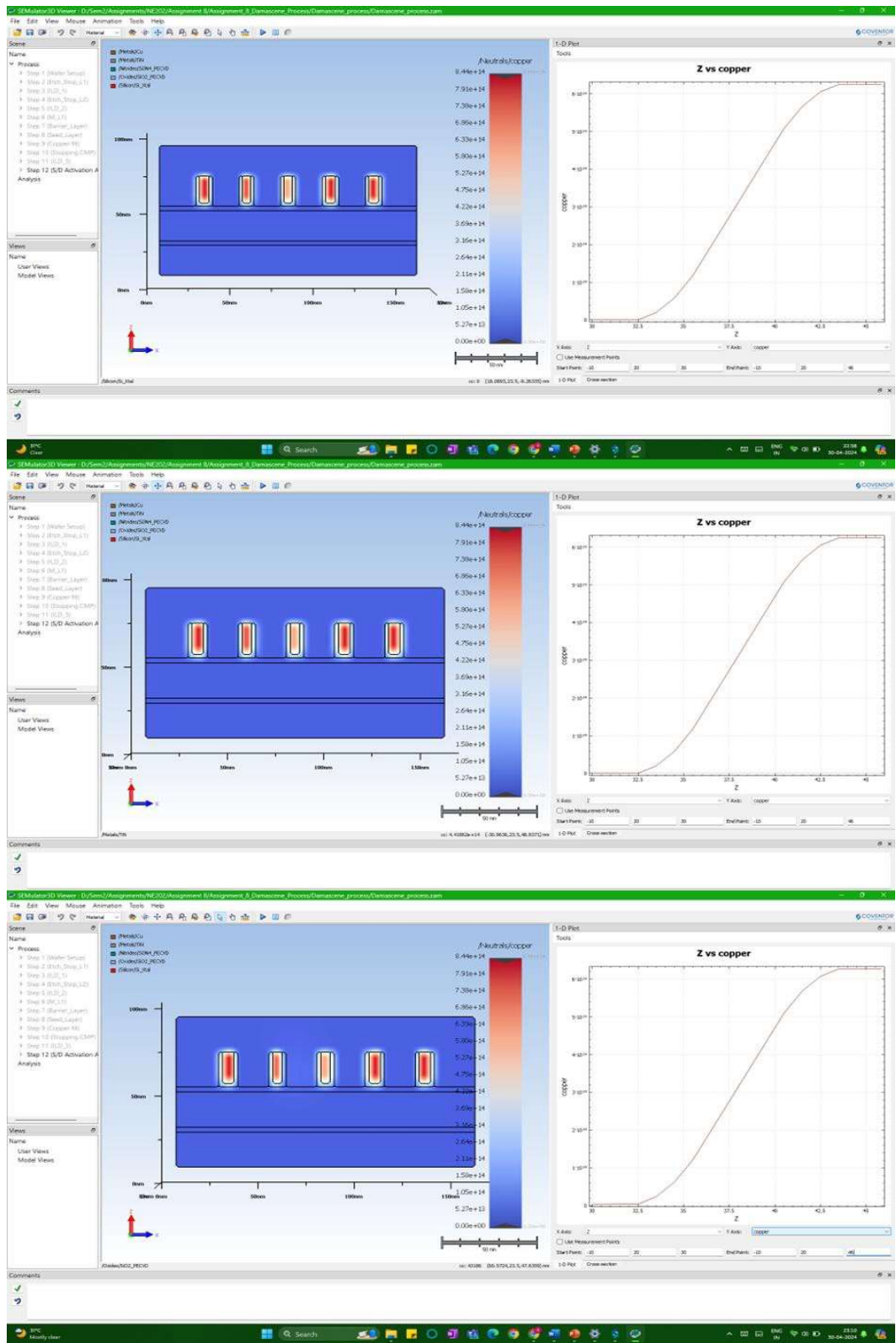


Figure 11 : Cu Profiles for TiN barrier

Simulated results for W barrier are shown in figure below,

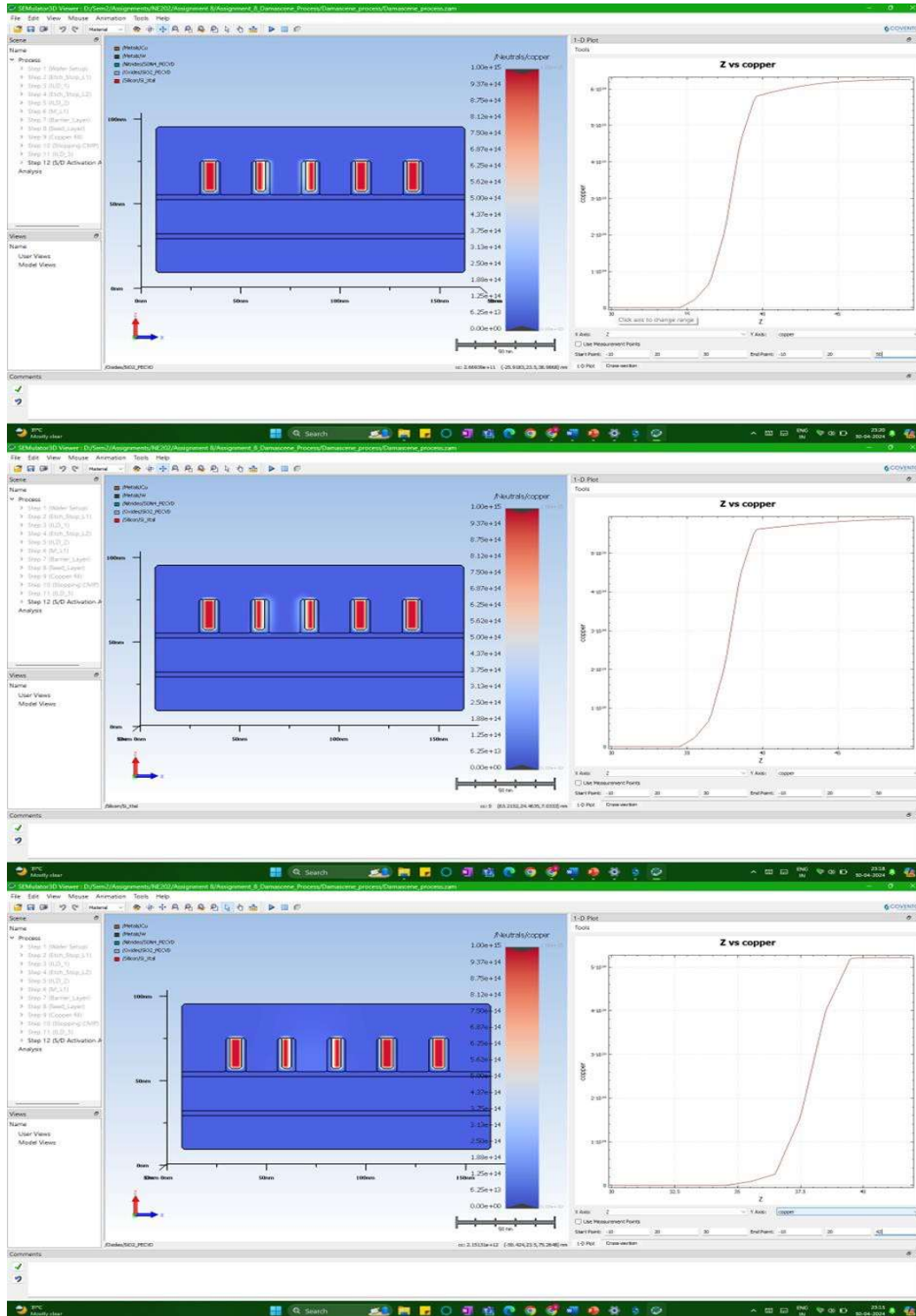


Figure 12 : Cu Profiles for W barrier

Among all barrier materials TiN seems to be best for stopping diffusion of Cu into junction at low anneal temperatures.

Question2(c)

The aspect ratio of a trench is critical in metal filling because techniques like sputtering, there is problem as atoms to be deposited need to follow a straight line but in high aspect ratio trenches, these atoms find it difficult to reach the bottom as the upper part or narrow opening of the trench can block the path, resulting in incomplete filling, uneven deposition, and sometimes void creation.

We perform copper electroplating over other deposition techniques to fill the trench because Copper electroplating overcomes these limitations for trench filling as electroplating uses electricity instead of line of sight to deposit copper. This deposits copper on all surfaces of the trench equally and conformally, regardless of the aspect ratio.

Question 3: 14nm advanced logic process

This task is centred around 14 nm advanced logic process, we aim to pinpoint the specific steps where damascene process is done. After that we try to identify the barrier material and etch stop layer.

Figure 13 shows process flow for this simulation.

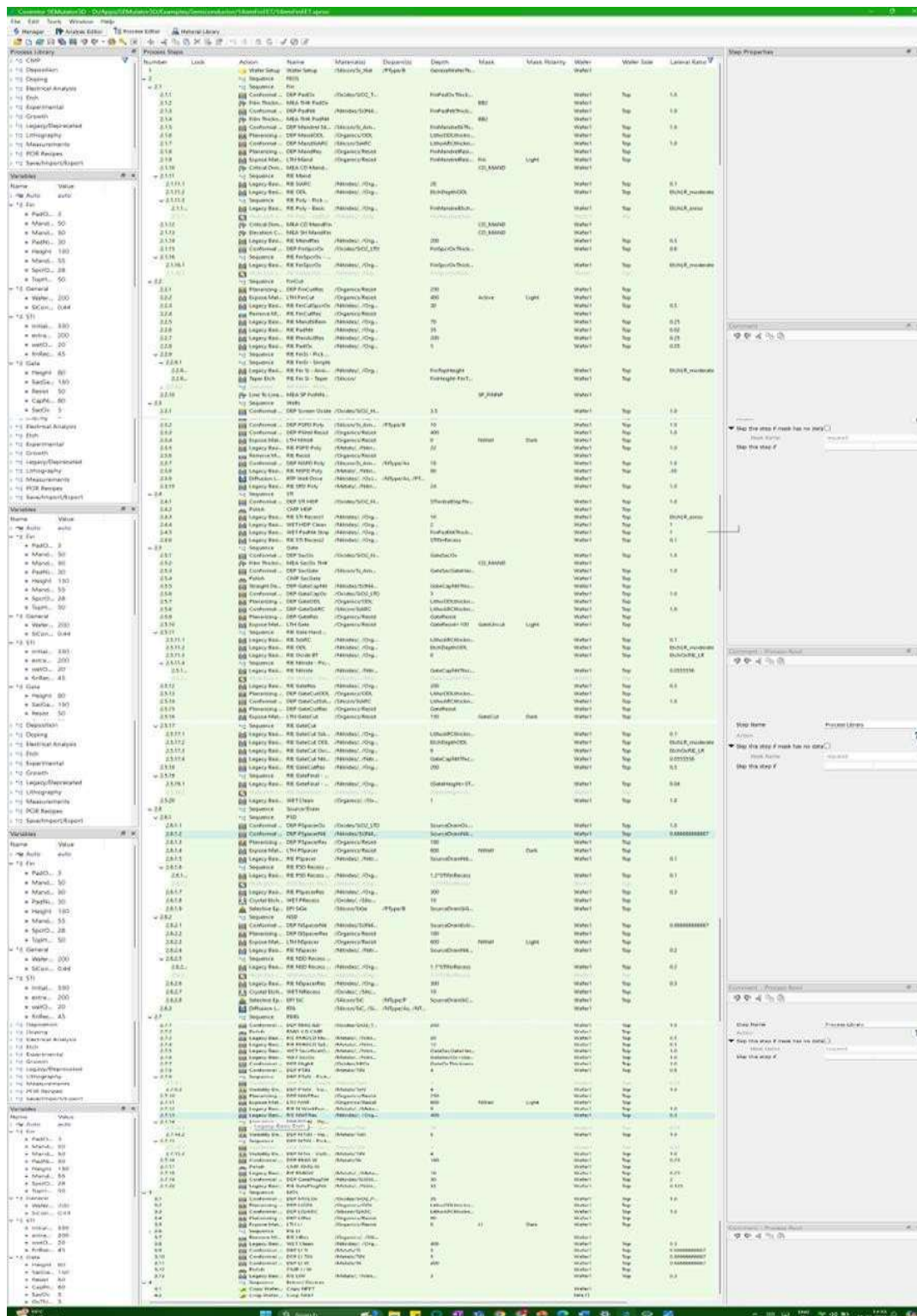


Figure 13 : Steps involved in FinFET simulation

Resultant structure is shown by Figure 14 and Figure 15.

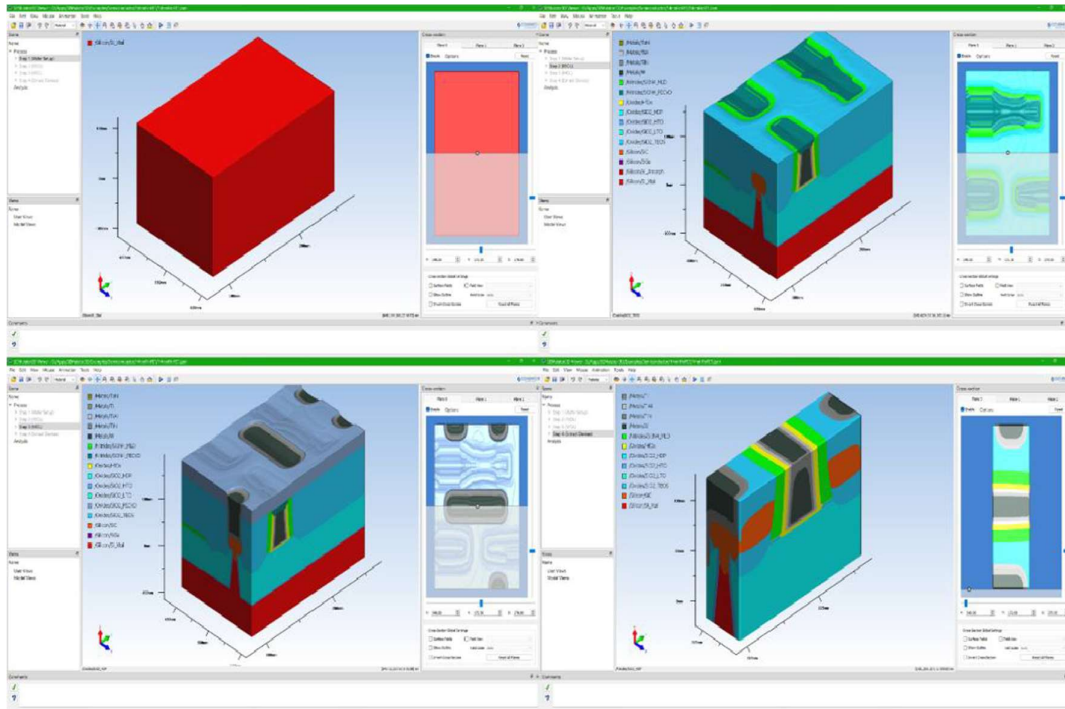


Figure 14 : Simulated 14nm FinFET process (3D view)

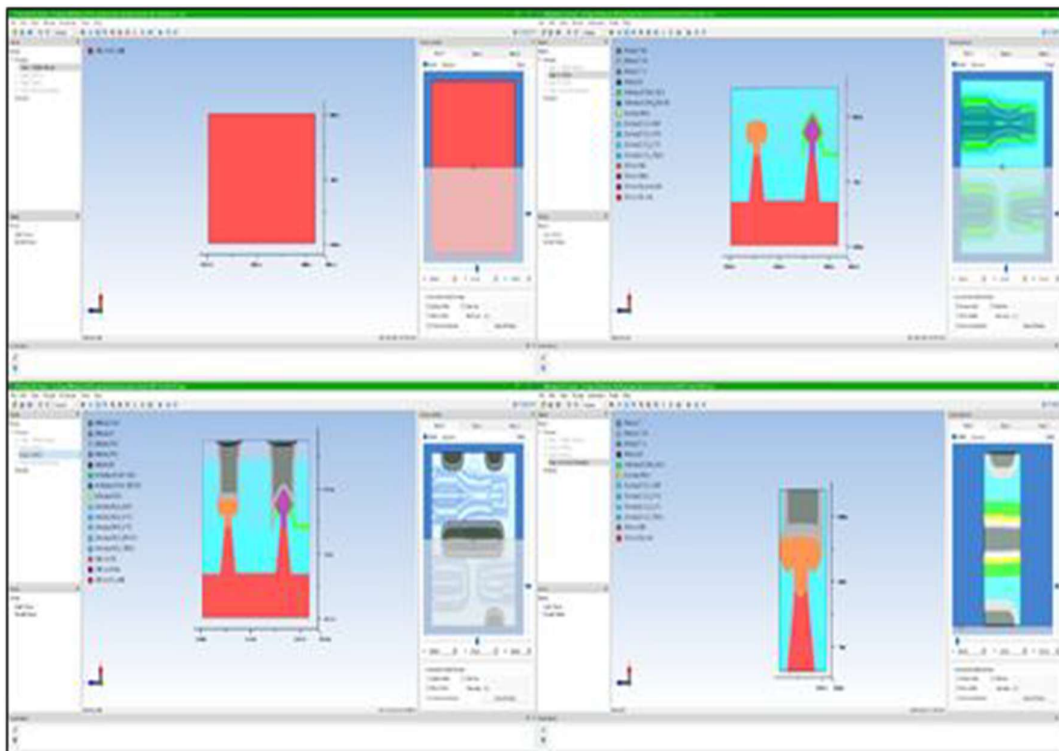


Figure 15 : Simulated 14nm FinFET process (Front View)

Question3(a)

Step 3.9 to Step 3.11 is Damascene process without seed layer deposition.

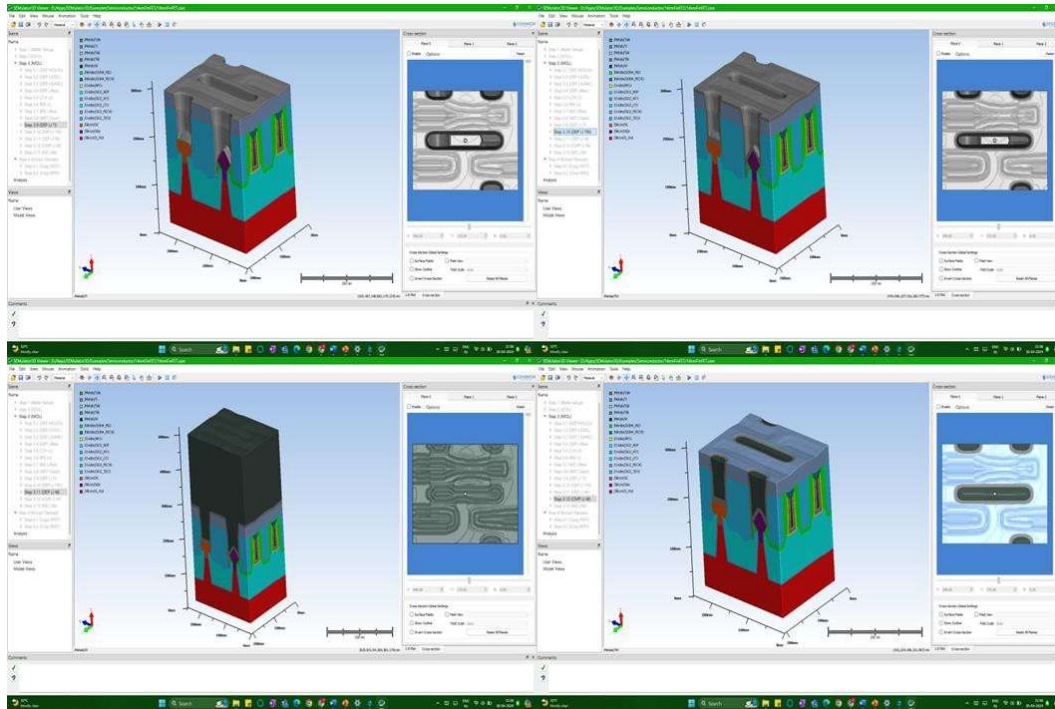


Figure 16 : Damascene Process in 14nm FinFET process

Question3(b)

Following layers are there in this process,

- Liner metal – Ti.
- Barrier metal – TiN.
- Metal 1 – Tungsten (W).
- Etch stop layer – SiO₂_PECVD.

Conclusion

We aim to understand how the damascene process addresses the limitations of conventional metallization. The damascene approach offers a more controlled and reliable method for fabricating copper interconnects in integrated circuits. Analysing diffusion profiles and the effectiveness of barrier layers provides valuable insights for optimizing the damascene process.

References

- [1] https://doi.org/10.1007/978-3-642-57121-3_3
- [2] [https://doi.org/10.1016/0040-6090\(82\)90429-1](https://doi.org/10.1016/0040-6090(82)90429-1)
- [3] [https://doi.org/10.1016/0254-0584\(95\)01514-0](https://doi.org/10.1016/0254-0584(95)01514-0)